

## 2-amaliy mashg'ulot. Raqamli tasvir: sensor, rang, siqish

2026–2027 o'quv yili

### Muhit (setup)

Python 3.11 yoki 3.12, faqat CPU. Versiyalar qat'iy belgilangan (`requirements.txt`): `numpy==2.4.4`, `scipy==1.17.1`, `pandas==3.0.2`, `scikit-learn==1.8.0`, `scikit-image==0.26.0`, `matplotlib==3.10.9`, `pillow==12.2.0`, `opencv-python-headless==4.13.0.92`, `torch==2.14.0`, `torchvision==0.29.0`.  
O'rnatish: `python -m venv .venv`, so'ng `pip install -r requirements.txt`. Import nomi: `import cv2`. Tasodifiylik: `np.random.default_rng(42)`. Barcha rasmlar `skimage.data` ichida (internet shart emas); natija rasmlari `./out/` ga saqlanadi.

**Mavzu.** Raqamli tasvir zanjiri: foton (Poisson) va o'qish shovqini, SNR, dinamik diapazon; diskretlash, Naykvist chegarasi, alyasing (aliasing); Bayer naqshi va demozaiklash (demosaiicing); kvantlash; gamma va sRGB; xotira (`dtype`, `strides`, HWC/CHW, BGR); `uint8` toshishi; rang fazolari (YCrCb, HSV, Lab); PNG/JPEG, DCT, xromatik subdiskretlash; sifat o'lchovlari MSE, PSNR, SSIM. **Maqsadlar.** Sensor shovqini va kvantlash xatosini hisoblash va simulyatsiya qilish; alyasing va gamma xatolarini ko'rsatish; massiv xotirasini va toshishni boshqarish; rang fazosi va siqish formatini asosli tanlash; PSNR/SSIM bilan sifatni o'lchash. **Baholash.** 1–16-topshiriqlar (qo'lda hisoblash va nazariya) — har biri 2 ball (32); 17–30-topshiriqlar (kod) — har biri 4 ball (56); kod sifati va hisobot — 12 ball. Jami 100 ball. Kod `python solutions.py` bilan xatosiz, takrorlanuvchan ishlashi shart.

**Belgilashlar.** `uint8` diapazoni  $L = 255$ ;  $\text{PSNR} = 10 \lg(255^2/\text{MSE})$ ; luma  $Y = 0,299R + 0,587G + 0,114B$ ; OpenCV (8 bit):  $C_r = 0,713(R - Y) + 128$ ,  $C_b = 0,564(B - Y) + 128$ , HSV da  $H/2 \in [0, 180)$ ,  $S, V \in [0, 255]$ ; soddalashtirilgan gamma  $V = L^{1/2,2}$ ; sRGB:  $L \leq 0,0031308$  da  $V = 12,92L$ , aks holda  $V = 1,055L^{1/2,4} - 0,055$ .

### 1. Qo'lda hisoblash, keltirib chiqarish va nazariy savollar

- SNR.**  $g = 1$ , o'qish shovqini  $\sigma_r = 6e^-$ , o'rtacha  $\mu = 400$  foto-elektron. SNR ni (marta va dB da,  $20 \lg \text{SNR}$ ) toping va sof foton chegarasi  $\sqrt{\mu}$  bilan solishtiring.
- Kerakli yorug'lik.**  $\sigma_r = 4$ ,  $g = 1$ .  $\text{SNR} = 30$  uchun  $\mu$  ni toping. O'qish shovqini bo'lmaganda-chi? Ekspozitsiya necha foizga uzayadi?
- Dinamik diapazon.** Qudug sig'imi (FWC)  $20\,000e^-$ ,  $\sigma_r = 4e^-$ . DR ni dB va "pog'ona"larda ( $\log_2$ ) toping. ADC kamida necha bitli bo'lishi kerak?
- Alyasing.** (a)  $f_s = 1 \text{ px}^{-1}$  da  $0,62$  sikl/px naqsh qaysi chastotada ko'rinadi? (b)  $0,3125$  sikl/px naqshli tasvirdan har 4-piksel olinsa, yangi tasvirda chastota qancha? (c) Xuddi shu kichraytirishda 4 ta qo'shni pikselni o'rtachalash amplitudani necha marta kamaytiradi ( $|\sum_{k=0}^3 e^{2\pi i \cdot 0,3125k}|/4$ )?
- Bayer va hajm.** RGGB naqshida R pikselining qo'shnilari:  $G$  (yuqori, past, chap, o'ng) = 100, 120, 90, 110;  $B$  (4 diagonal) = 40, 60, 50, 70. Bilinear  $\hat{G}$ ,  $\hat{B}$  ni toping.  $6000 \times 4000$  sensor, 14 bitli RAW kadr necha bayt? Demozaiklangan 8 bitli RGB va YCbCr 4:2:0 kadr-chi? Qiymatlarning qancha qismi taxmin?
- Kvantlash.** 5 bitli kvantlash ( $[0, 256)$  oralig'ida):  $q$ ,  $\text{Var } e = q^2/12$ , PSNR ni aniq formula va  $6,02n + 10,76$  bilan toping.
- Keltirib chiqarish.**  $e \sim U[-q/2, q/2]$  uchun  $\text{Var } e = q^2/12$  ni va  $\text{PSNR} \approx 6,02n + 10,76 \text{ dB}$  ni isbotlang. Formula qachon ishlamaydi (ikki holat)?
- Gamma kodlar.**  $L \in [0, 0,05]$  oraliqqa chiziqli ( $\text{round}(255L)$ ) va gamma ( $\text{round}(255L^{1/2,2})$ ) kodlashda nechta kod tegadi?  $L = 0,002$  va  $L = 0,2$  ning aniq sRGB kodini toping.
- Gamma-to'g'ri o'rtacha.**  $\gamma = 2,2$ . Kodlari 60 va 220 bo'lgan piksellarning to'g'ri (chiziqli fazodagi) o'rtachasini toping; naiv natija bilan farqi qancha?
- Strides.**  $720 \times 1280 \times 3$  `uint8` (C-tartib) massivda `strides`,  $[100, 200, 2]$  elementning bayt siljishi, massiv hajmi; `float32` ga o'tkazilgandagi `strides`. `transpose(2, 0, 1)` dan keyingi `strides` va nusxa olinadimi?
- Toshish.** `uint8`:  $a = [230, 15, 128]$ ,  $b = [40, 20, 200]$ . NumPy da  $a + b$ ,  $a - b$ ,  $b - a$  va `cv2.add`, `cv2.subtract` natijalari.
- Rang fazolari.** RGB = (50, 120, 200) pikselni OpenCV ning YCrCb va HSV (8 bit) qiymatlariga o'tkazing (yaxlitlab).
- Doiraviy o'rtacha.** Uch pikselning OpenCV  $H$  qiymatlari 170, 5, 10. Ularni gradusga o'tkazib, oddiy va doiraviy ( $\text{atan2}(\sum \sin, \sum \cos)$ ) o'rtachani toping. Qaysi biri to'g'ri va nega?
- DCT.** Barcha qiymatlari 150 bo'lgan  $8 \times 8$  blok:  $f = 150 - 128$ ;  $F(0, 0)$  ni va  $Q(0, 0) = 16$  da kvantlangan

koeffitsiyentni toping. Ortonormal DCT da  $F(0,0) = 8\bar{f}$  ekanini ko'rsating.

15. **Interpolyatsiya.**  $I = \begin{pmatrix} 40 & 80 \\ 120 & 200 \end{pmatrix}$ ,  $(x,y) = (0,3; 0,7)$  ( $x$  — ustun). Bilinear va eng yaqin qo'shni (nearest) qiymatlarini toping. Segmentatsiya niqobini qaysi usul bilan masshtablash kerak va nega?
16. **PSNR va SSIM.** (a) MSE = 50 da PSNR. (b) Bitta oyna:  $\mathbf{x} = (90, 100, 110, 120)$ ,  $\mathbf{y} = \mathbf{x} + 20$ ,  $C_1 = (0,01 \cdot 255)^2$ ,  $C_2 = (0,03 \cdot 255)^2$ : SSIM va PSNR ni toping. Nima uchun ular turlicha baho beradi?

## 2. Dasturlash topshiriqlari

Barcha kodlar bitta `solutions.py` faylida: har bir topshiriq — alohida funksiya (`task17()`, ..., `task30()`) va `main()`. Umumiy importlar:

```
import numpy as np, cv2, torch, skimage.data
from scipy import ndimage
from skimage.metrics import peak_signal_noise_ratio as psnr, structural_similarity as ssim
rng = np.random.default_rng(42) # har bir tasodifiy topshiriq boshida yangidan
# skimage.data.astronaut(): (512,512,3) RGB uint8; skimage.data.camera(): (512,512) uint8
```

Hujjatlar: <https://scikit-image.org/docs/stable/api/skimage.data.html>, <https://scikit-image.org/docs/stable/api/skimage.metrics.html>, [https://docs.opencv.org/4.x/d8/d01/group\\_imgproc\\_\\_color\\_\\_conversions.html](https://docs.opencv.org/4.x/d8/d01/group_imgproc__color__conversions.html), [https://docs.opencv.org/4.x/da/d54/group\\_imgproc\\_\\_transform.html](https://docs.opencv.org/4.x/da/d54/group_imgproc__transform.html), [https://docs.opencv.org/4.x/d4/da8/group\\_imgcodecs.html](https://docs.opencv.org/4.x/d4/da8/group_imgcodecs.html), <https://numpy.org/doc/stable/reference/arrays.ndarray.html>.

17. **Sensor shovqini modeli.** `sensor_samples(mu, sigma_r, n, rng, g=1.0) = g*rng.poisson(mu,n)+rng.normal(0,sigma_r,n)`.  $\sigma_r = 4$ ,  $\mu \in \{5, 50, 500, 5000\}$  (shu tartibda, bitta `rng`),  $n = 100\,000$ . Jadval: o'lchangan dispersiya,  $\mu + \sigma_r^2$ , o'lchangan SNR (`mean/std`), nazariy SNR,  $\sqrt{\mu}$ .
18. **Photon transfer curve (real kalibrlash usuli).** `photon_transfer(g, sigma_r, mus, rng)`: har bir  $\mu$  uchun ikki  $200 \times 200$  tekis kadr  $a, b = g \cdot \text{Poisson}(\mu) + g \cdot \mathcal{N}(0, \sigma_r^2)$ ; o'rtacha  $=(\bar{a} + \bar{b})/2$ , dispersiya  $= \text{Var}(a - b)/2$ .  $g = 0,5 \text{ DN}/e^-$ ,  $\sigma_r = 3$ ,  $\mu \in \{20, 50, 100, 200, 400, 800, 1600\}$ . `np.polyfit(mean, var, 1)`: qiyalik  $= g$ ,  $\sigma_r = \sqrt{\text{kesma}/g}$ . 12 bitli ADC ning 4095 DN siga necha elektron mos keladi?
19. **Alyasing.**  $512 \times 512$  vertikal chiziqlar: `127.5+100*cos(2*pi*0.3125*n)`. `dominant_freq(row)` — `np.fft.rfft` bo'yicha eng kuchli (DC siz) chastota va amplituda. Original, `img[:4,:4]` va `cv2.resize(..., (128,128), interpolation=cv2.INTER_AREA)` uchun ( $f, A$ ); nazariy amplituda bilan (4(c)) solishtiring.
20. **Bayer va demozaiklash.** `bayer_rrgb(rgb)` — `astronaut()` dan RGBB mozaika; `cv2.cvtColor(m, cv2.COLOR_BayerBG2RGB)` (to'g'ri kod) va `COLOR_BayerRG2RGB` (noto'g'ri); `demosaic_bilinear(m)` — o'zingizniki: niqoblangan kanal \* yadro (`ndimage.convolve, mode='mirror'`), R/B uchun  $\frac{1}{4} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$ , G uchun  $\frac{1}{4} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ . Uchala PSNR, ichki sohada (`[2:-2]`) OpenCV bilan maksimal farq, eng katta xatoli piksel; `out/t20_demosaic_crop.png` (`[180:308,150:278]`).
21. **Kvantlash jadvali.** `quantize(img, n) = q[x/q] + q/2`,  $q = 256/2^n$ . `camera()` uchun  $n = 1..7$ : Var  $e$ ,  $q^2/12$ ,  $\bar{e}$ , PSNR, nazariy PSNR. Farazlar qayerda buziladi?
22. **sRGB.** `srgb_encode(L)`, `srgb_decode(V)` (vektorlashgan, `np.where`). (a) 0.255 kodlari uchun `decode`→`encode`→`round` xatosi; (b)  $L \in [0;0,1]$  dagi kodlar soni (chiziqli va sRGB, `np.linspace(0,0.1,100001)`); (c)  $256 \times 256$  bir piksellik 0/255 shaxmatni `INTER_AREA` bilan  $128 \times 128$  ga naiv va chiziqli fazoda kichraytirish (o'rtacha qiymat).
23. **Xotira.** `astronaut()` ni BGR ga o'tkazish: `shape, strides, [100,200,1]` siljishi va `ravel()` bilan tekshiruv; CHW (`transpose`) `strides, np.shares_memory, C_CONTIGUOUS; float32` `strides` va `nbytes`; `torch.from_numpy(np.ascontiguousarray(chw))` ning `stride()`; 10 000 ta  $224 \times 224 \times 3$  to'plam hajmi (GB).
24. **Toshish va saqlash.** 11-topshiriqni kodda tekshirish (massivlar shakli (1,3)). `astronaut()*1.8`: 255 dan oshganlar soni, o'rtacha: original, `uint8` toshishi (`astype(int64)%256`), `to_uint8(x) = round(clip(x,0,255))`. `camera()*0.7`: aniq o'rtacha, `astype(np.uint8)` va `round` o'rtachasi.
25. **Rang fazolari.** `rgb_to_ycrcb_hsv(r,g,b)` — formulalar bo'yicha (float). RGB (200,100,50) va (50,120,200) uchun `cv2.cvtColor` bilan solishtiring. `astronaut` da BGR→YCrCb/HSV/Lab→BGR (`uint8`) aylanma xatosi: maksimal va o'rtacha.
26. **Rang bo'yicha segmentatsiya (amaliy).** `astronaut` (BGR→HSV); qizil-to'q sariq soha: `cv2.inRange(hsv, (0,120,70), (10,255,255))`  $\cup$  `inRange(hsv, (170,120,70), (179,255,255))`,  $3 \times 3$  MORPH\_OPEN; `cv2.connectedComponentsWithStats`: piksellar soni, komponentalar soni, eng katta komponenta maydoni va bbox. Niqobdagi  $H$  (gradusda,  $2H$ ) ning oddiy va doiraviy o'rtachasi (`circular_mean_deg`). `out/t26_red_mask.png`.
27. **PNG va JPEG.** `astronaut` (BGR): `cv2.imencode('.png')` hajmi, siqish koeffitsiyenti, yo'qotishsizlik; JPEG sifat  $q \in \{95, 80, 50, 25, 10\}$  (`IMWRITE_JPEG_QUALITY`): bayt, siqish, PSNR (rangli), SSIM (kulrang,

`data_range=255`). Sifat qiymatini tavsiya qiling.

- 28. Niqobni saqlash (amaliy xato).** `mask=np.digitize(camera(), [50,100,150,200])` (sinflar 0..4, `uint8`). PNG va JPEG ( $q = 90$ ) orqali saqlab-oʻqing: oʻzgargan yorliqlar %, unikal qiymatlar.  $256 \times 256$  ga `INTER_NEAREST` va `INTER_LINEAR` bilan kichraytirish: `INTER_NEAREST` natijasidan farq qiluvchi (aralash yorliqli) piksellar %.
- 29. Xromatik subdiskretlash.** `subsample_channels(bgr, idx)`: YCrCb da `idx` kanallarni `INTER_AREA` bilan 2 marta kichraytirib, `INTER_LINEAR` bilan qaytaring. ( $C_r, C_b$ ) va faqat  $Y$  uchun PSNR;  $1920 \times 1080$  kadr 4:4:4 va 4:2:0 da necha bayt?
- 30. JPEG yadrosi: DCT va kvantlash.** `dct_matrix(8)` — ortonormal DCT-II ( $F = DfD^T$ ); `cv2.dct` bilan solishtiring. `jpeg_like(gray, scale)`:  $8 \times 8$  bloklar,  $-128$ , DCT,  $\text{round}(F/(s \cdot Q_Y))$ , teskari DCT, `to_uint8`;  $Q_Y$  — standart JPEG yorqinlik jadvali (ITU T.81, K ilova):

```
QY = np.array([[16,11,10,16,24,40,51,61], [12,12,14,19,26,58,60,55], [14,13,16,24,40,57,69,56],
               [14,17,22,29,51,87,80,62], [18,22,37,56,68,109,103,77], [24,35,55,64,81,104,113,92],
               [49,64,78,87,103,121,120,101], [72,92,95,98,112,100,103,99]], float)
```

`camera()` uchun  $s \in \{0,5;1;2;4\}$ : nol koeffitsiyentlar %, PSNR, SSIM; `out/t30_jpeg_like_s4.png`.